

R2: Spectral Shape

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Where should the wave energy lie for the inverse to work best?

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SHARC buoy project Code: `runR2SpectralShape.m`, `buildR2Seas.m` Results: `Results/R2/R2_20261003_130557`

Summary

Question. With the floe (p_0), layout (Level 2C, reference s26), noise (LSM6DSV16X), record ($N = 65536$) and the wave energy inside the retained band all fixed, how does the distribution of that energy across frequency change precision and Welch bias? 16 pre-registered seas: JONSWAP with $\omega_p \in \{3.5, 4.25, 5, 6, 7.5\}$ rad/s and $\gamma_J \in \{1, 3.3, 7\}$, plus one bimodal sea, each at $L = 1024, 2048, 4096$. **Outcome.** The exact transmissibility and its Jacobian are identical for every sea (difference 0), yet the practical inverse depends strongly on the sea: at $L = 2048$ $\sigma_{\ln \beta}$ ranges from 0.161 to 0.766% (a factor 4.8 at equal in-band energy) and d_W from 0.048 to 0.359. **All 16 seas are Identifiable at $L = 2048$** , the shortest tested length that is adequate for every shape ($L = 1024$: 3 of 16). Energy in the upper part of the band gives the best precision; a narrow sea at the lower band edge gives the worst. $L = 4096$ gives the smallest RMSE in $\ln \beta$ for all 16 seas. M1 case (pre-fixed rule): the narrow low-frequency sea, $\omega_p = 3.5$ rad/s, $\gamma_J = 7$.

1 What the study does

Controlled experiment. Only the sea and L change. Fixed: $p = p_0$, Level 2C with reference s26, LSM6DSV16X noise, $N = 65536$, $\Delta t = 0.1$ s, the retained band 3.178 to 8.301 rad/s, the cached EMM field, the estimator and the class thresholds (checked: tester 10/10, the reference sea reproduces `evaluateScenario` bitwise). The bimodal sea has peaks at 3.5 and 6.5 rad/s ($\gamma_J = 7$ and 3.3) carrying 40 and 60% of the in-band variance.

Fixed in-band variance (a logged clarification of the programme’s “fixed H_s ”). Every sea satisfies

$$\int_{3.178}^{8.301} S_{\eta}^{(q)}(\omega) d\omega = \int_{3.178}^{8.301} S_{\eta, \text{ref}}(\omega) d\omega = 1.407 \times 10^{-4} \text{ m}^2,$$

so R2 changes *where* the usable energy lies, not *how much* there is (R1 covers the level). The single-peak seas therefore have H_s from 0.049 to 0.072 m; the reference keeps exactly $H_s = 0.05$ m.

Why the shape can matter at all. For one incident component $S_{jr} = H_j H_r^* S_{uu}$ and $S_{rr} = |H_r|^2 S_{uu}$, so $T_j = S_{jr}/S_{rr} = H_j/H_r$: the exact transmissibility and its Jacobian J do not depend on S_{uu} . Two practical quantities do. (i) The covariance Σ_k depends on the per-bin signal-to-noise ratio, so the sea decides which bins carry information in $F = \sum_k J_k^T \Sigma_k^{-1} J_k$. (ii) The Welch estimate smooths before taking the ratio, $\hat{T}_j \approx (W * S_{jr}) / (W * S_{rr})$, so where T varies across the window W , the result depends on how S_{uu} weights the neighbouring frequencies: the Welch bias depends on the sea.

2 Results

2.1 Production setting, $L = 2048$

ω_p (rad/s)	$\sigma_{\ln \beta}$			d_W		
	$\gamma_J = 1$	3.3	7	$\gamma_J = 1$	3.3	7
3.5	0.493%	0.624%	0.766%	0.263	0.238	0.221
4.25	0.377%	0.469%	0.566%	0.261	0.272	0.283
5	0.292%	0.355% [†]	0.419%	0.244	0.309 [†]	0.359
6	0.225%	0.251%	0.272%	0.184	0.248	0.287
7.5	0.176%	0.165%	0.161%	0.065	0.049	0.048
bimodal 3.5 + 6.5	0.264%	$(\sigma_{\ln R} 0.052\%, \kappa 67)$		0.170		

[†]Reference sea (= E1/ p_0). $\sigma_{\ln R}$ follows $\sigma_{\ln \beta}$ (0.032 to 0.141%); $\kappa(F)$ 49 to 120. All 16 Identifiable.

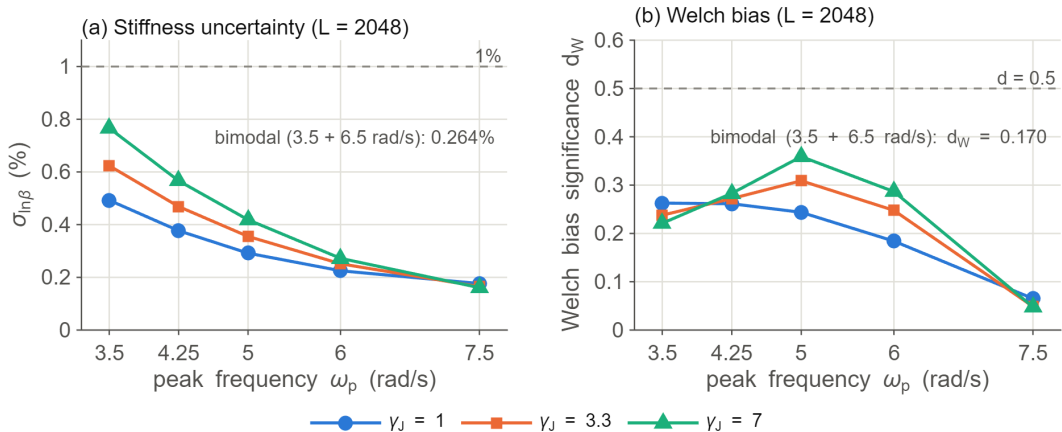


Figure 1. The 15 single-peak seas at $L = 2048$: (a) stiffness uncertainty and (b) Welch bias significance against the peak frequency, one line per peakedness. The bimodal sea has no single peak frequency and is annotated. Dashed: the Identifiable limits.

Peak location dominates. For every γ_J , moving the peak from 3.5 to 7.5 rad/s lowers $\sigma_{\ln \beta}$ by a factor 2.8 to 4.8, monotonically. **Peakedness magnifies placement.** At $\omega_p = 3.5$ a narrower sea is worse (0.493, 0.624, 0.766% for $\gamma_J = 1, 3.3, 7$), at 5 rad/s likewise but less (0.292 to 0.419%), and at 7.5 rad/s the order reverses slightly (0.176, 0.165, 0.161%). **The Welch bias behaves differently:** it is not monotonic in ω_p , is largest near the centre of the band for the narrower seas (0.359 at $\omega_p = 5$, $\gamma_J = 7$) and smallest at 7.5 rad/s (about 0.05). Precision and bias are governed by related but distinct spectral mechanisms. The bimodal sea is better than the reference in both (0.264% against 0.355%, d_W 0.170 against 0.309).

2.2 Exact inverse versus finite estimator

Across all 16 seas $\max_q |f_0^{(q)} - f_{0,\text{ref}}| = 0$ and $\max_q |J^{(q)} - J_{\text{ref}}| = 0$: the physical inverse map does not change at all. Everything in Figure 1 comes from the experiment and the estimator.

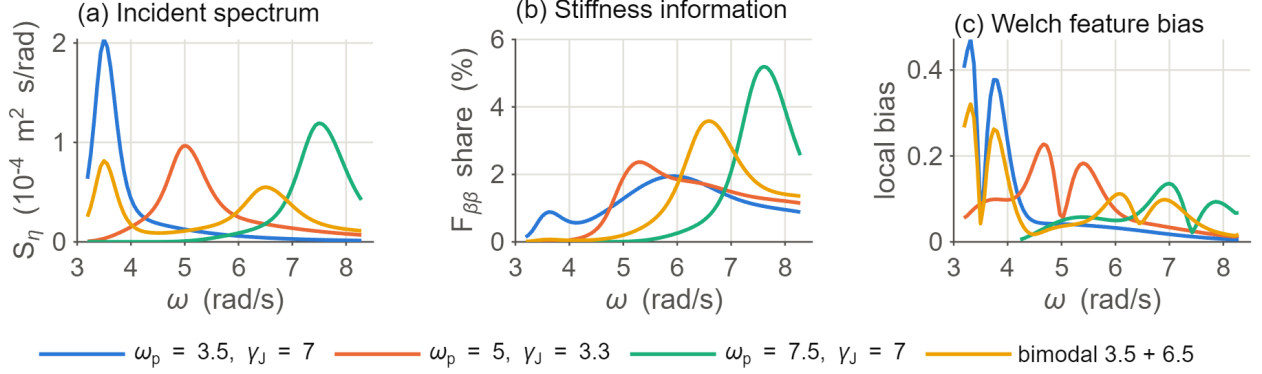


Figure 3. Mechanism at $L = 2048$ for four seas of equal in-band variance. (a) Incident spectrum on the retained bins. (b) Share of the diagonal Fisher information $F_{\beta\beta,k}$ per accepted bin; a frequency-local diagnostic, not a decomposition of $\sigma_{\ln \beta}$, which also depends on $F_{\beta R}$ and F_{RR} . (c) Local feature-bias significance $\sqrt{b_{W,k}^T \Sigma_k^{-1} b_{W,k}}$ per bin; a diagnostic, not a decomposition of d_W (the global bias is projected through F and bin contributions can cancel).

(b) The upper band is worth more per unit energy. For $\omega_p = 7.5$ almost all of $F_{\beta\beta}$ comes from above 6 rad/s, where its energy is. The low-frequency narrow sea is the telling case: its own peak at 3.5 rad/s contributes only a small bump, and most of its $F_{\beta\beta}$ comes from the weak spectral tail between about 5 and 7 rad/s. Energy near the lower band edge is spent in bins that say little about stiffness, so shifting the same energy upward lowers $\sigma_{\ln \beta}$.

(c) The Welch bias sits on the flanks of the spectral peak. Each sea's local bias peaks either side of its spectral maximum (about 3.3 and 3.8 rad/s for $\omega_p = 3.5$; 4.7 and 5.4 for the reference; 7.0 and 7.9 for $\omega_p = 7.5$), where the window averages over a steeply varying S_{uu} before the ratio is formed. For the $\omega_p = 7.5$ sea the curve is absent below about 4.2 rad/s: those bins carry so little signal that the estimator's denominator rule rejects them (`R2_sweep.csv`, bins used). Plausibly, mid-band peaks put these flanks on the bins that carry most of $F_{\beta\beta}$, which would explain the maximum of d_W near $\omega_p = 5$; only the global d_W classifies.

2.3 Welch length

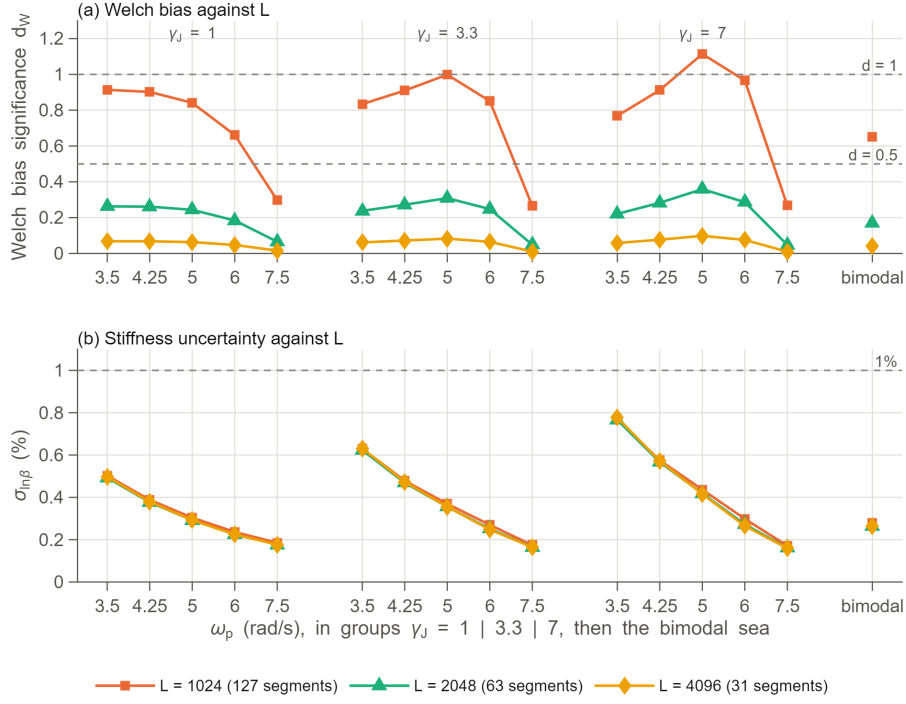


Figure 2. All 16 seas at $L = 1024, 2048, 4096$ ($N = 65536$: 127, 63, 31 segments): (a) Welch bias significance, (b) stiffness uncertainty. Colours as in D2.

L	Identifiable	Marginal	Not identifiable	d_W range
1024	3 ($\omega_p = 7.5$ only)	12	1 ($\omega_p = 5, \gamma_J = 7, d_W = 1.11$)	0.27 to 1.11
2048	16	0	0	0.05 to 0.36
4096	16	0	0	0.01 to 0.10

$\sigma_{\ln \beta}$ barely depends on L (Figure 2b); the bias falls by roughly a factor 4 per doubling of L , as in D2. With the RMSE $\sqrt{\sigma_{\ln \beta}^2 + \delta\theta_{W,\beta}^2}$, $L = 4096$ is best for all 16 seas at $N = 65536$.

3 Interpretation

- Where the energy lies, not how much, sets much of the precision.** Equal in-band variance does not mean equal information: redistributing the same energy changes $\sigma_{\ln \beta}$ by a factor 4.8.
- The upper part of the band is more informative for this configuration.** This holds for p_0 , Level 2C and the basin band; it is not a general claim that higher frequencies are always better. A1 showed that kR , which changes with the floe, organises the precision.
- Narrowness is neither good nor bad in itself.** It removes frequency diversity, so it amplifies whether the peak sits in an informative region: worst at 3.5 rad/s, best at 7.5 rad/s.
- A bimodal sea is benign here.** It is more precise and less biased than the reference; what matters is how its energy overlaps the informative bins, not the number of peaks.
- The physics is sea-independent; the experiment and the estimator are not.** T and J are unchanged; F , σ and d_W change. This separates the identifiability of the forward model from the information a given sea supplies and the bias a given estimator adds.
- $L = 2048$ passes the robustness test it was set:** 16 of 16 Identifiable, and it is the shortest tested length that is.
- $L = 4096$ is the RMSE optimum for a full 109 min record,** as D2 suggested. It also supports the A1 hypothesis that a longer L would recover the large- kR floes, which remains to be tested as a logged diagnostic.
- M1 stress case:** $\omega_p = 3.5, \gamma_J = 7, q = 0.766$. Its binding criterion is precision ($\sigma_{\ln \beta}/1\% = 0.766$ against $d_W/0.5 = 0.442$), so M1 probes weak Fisher information rather than Welch bias.

Limitations. One floe and one layout (p_0 , Level 2C), not the A1 domain. Fixed in-band variance rather than fixed H_s , deliberately (H_s varies between seas). One noise case (R1 and R3 cover the noise). The JONSWAP family plus one bimodal sea, one incident direction (V2, D3). Analytic expectations, not realised records (M1 tests the selected sea).

Requirement for the experiment and the thesis

For p_0 and Level 2C, the inverse stays Identifiable over the full tested range of peak frequency and peakedness with $L = 2048$ for a 109 min record; $L = 1024$ is not adequate. When stiffness precision matters, the excitation should place its energy toward the upper part of the retained band: concentrating it near the lower edge degrades $\sigma_{\ln \beta}$ by up to a factor 4.8 at the same usable energy. The sea's spectral shape is therefore a design variable, not only an amplitude condition. **M1 hand-off:** $\omega_p = 3.5$ rad/s, $\gamma_J = 7$.

Outputs (Results/R2/): R2_20261003_130557.log/.mat/_summary.txt; R2_sweep.csv, R2_L.comparison.csv, R2_M1.selection.csv; figures/R2_fig1 to R2_fig3 (.pdf/.png).